

What Is in a Flow?

Real-Time Filtering of Mixed Stock–Flow Systems.

A Technical Note^{*}

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Abstract

What is in a measured flow? Exactly four terms: the carry-over of the state from the window’s opening, the path news the shocks generate within it, the window’s structural increments themselves—the pass-through—and the release error. This note establishes the filtering machinery that takes the third term seriously. In continuous time the economy imprints its structural increments on measured flows as they occur; discrete-time treatments integrate that content away. The pass-through’s fingerprint is a flow–state cross-covariance, carried by a correlated-noise Kalman–Bucy filter and read back online by fixed projection operators with attainable ceilings in closed form. The filter delivers, at each release, both objects of real-time interest: the latent state and the contemporaneous structural shocks. The route is complementary to the structural VAR tradition, in which the invertibility condition returns a binary verdict rather than a computable ceiling. The framework is stated once, with proofs: the two measurement primitives for a reported flow and their exact discrete filters, the two-step sampling algorithm for arbitrary release calendars, well-posedness up to the stochastic-singularity boundary, and verifiable assessment criteria. Simulation evidence is developed in Rátfai (2026), the deployment on US data in Parra-Alvarez et al. (2026).

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1 Introduction

What is in a measured flow—and what can a filter that represents flows deliver as data arrive? This note gives the technical answer: it establishes the fundamentals of the online tracker developed in Rátfai (2026)—hereafter the companion paper—the machinery through which a fully structural continuous-time economy is tracked in real time. Two

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commitments distinguish the exercise; they are recorded here technically, their philosophy and economic case being the subject of the companion paper. First, both objects of real-time interest are delivered contemporaneously from the same structural model: the latent state—the nowcasting object—and the structural shocks hitting the economy, projected from the arriving release by a fixed operator. The route is complementary to the structural VAR tradition, and the two serve different uses: retrospective identification under structure imposed at the identification step alone, and real-time delivery under a full structural commitment—with the binary invertibility question of the former (Fernández-Villaverde et al., 2007) returning, under the latter, as a computable ceiling on attainable recovery. Second, the machinery exploits information retained within the observation interval: in continuous time the structural increments enter measured flows as they occur—the pass-through—and the correlated-noise filter reads that content rather than integrating it away.

The construction extends the exact-discretisation tradition of continuous-time econometrics (Bergstrom, 1986; Harvey and Stock, 1993; Zdrozny, 1988; Christensen et al., 2024) in two respects: from parameter estimation to real-time filtering, and in the object discretised—the same Van Loan machinery the tradition applies to the integral-form flow (Primitive I) applies directly to the rate-form primitive (Primitive II), whose exact discrete filter is also constructed here. The framework is stated once, in full generality— n -dimensional state, q structural shocks, an arbitrary partition of m observables into stocks and flows—with the associated proofs, algorithms, the online tracking criterion, well-posedness results, the information bounds of sampling, and assessment criteria collected in one place, so that applications can cite the machinery without rederiving it. The exposition is deliberately equation-forward: each section states its objects, the conditions under which they exist, and the identities connecting them. No calibrated model and no simulation results appear here—though several identities below are checked numerically where stated; the model-specific factors that organise filter comparisons (Φ , Π , Ψ_I , Ψ_{II} , Λ , and the recovery-correlation law), and all quantitative results, are the subject of the companion paper, and the deployment of the framework on observed macroeconomic data is the subject of Parra-Alvarez et al. (2026).

2 The System and the Two Measurement Primitives

What is in a measured flow? Exactly four terms: the carry-over of the state from the window’s opening, the path news the shocks generate within the window, the window’s structural increments themselves—the pass-through—and the release error; Proposition 2.4 states the decomposition exactly. The two measurement primitives of this section differ in precisely one of the four, the pass-through, and the flow–state cross-covariance it generates is the organising object of the framework. The route runs from the model to the data: system, primitives, release convention, exact window decomposition, collected ABCD form, and the two channels through which stocks and flows convey the contemporaneous increment.

Notation. $x_t \in \mathbb{R}^n$ is the latent state; $W_t \in \mathbb{R}^q$ a standard Brownian motion with independent components; releases arrive on the grid $t_k = kh$, $h > 0$. For $\tau \geq 0$ define the

flow operator and its integral

$$\Gamma(\tau) = \int_0^\tau e^{Au} du, \quad \Gamma_h \equiv \Gamma(h), \quad V_h \equiv \int_0^h \Gamma(\tau) d\tau, \quad (1)$$

with $\Gamma(\tau) = A^{-1}(e^{A\tau} - I)$ and $V_h = A^{-1}(\Gamma_h - hI)$ whenever A is invertible; the integral definitions in (1) remain valid for singular A . $\Omega = BB'$ denotes the process-noise variance rate (variance per unit time; the *noise intensity* of the filtering literature), I_p the $p \times p$ identity, and $\text{diag}(v)$ the diagonal matrix built from a vector v . Transposition is $(\cdot)'$. Two conventions on η are used throughout: a *superscripted* η is a window innovation— η_k^x for the state, η_k^f for the flow—while a *subscripted* η is a release-error scale, η_f on the flow channels, η_s on the stocks, η_j on a generic channel j . The release errors themselves are written ξ , reserving ε for the structural innovations.

2.1 Continuous-time system and observation partition

The state follows the linear stochastic differential equation

$$dx_t = Ax_t dt + B dW_t, \quad A \in \mathbb{R}^{n \times n}, \quad B \in \mathbb{R}^{n \times q}, \quad (2)$$

and the contemporaneous signal vector is the level $y_t = Cx_t$ with $C \in \mathbb{R}^{m \times n}$. The m observables partition into m_f flows and m_s stocks, $m = m_f + m_s$, with loadings stacked conformably,

$$C = \begin{bmatrix} C^f \\ C^s \end{bmatrix}, \quad C^f \in \mathbb{R}^{m_f \times n}, \quad C^s \in \mathbb{R}^{m_s \times n}. \quad (3)$$

All randomness in the system is Gaussian—Brownian state innovations and Gaussian release errors—so states and observables are jointly Gaussian and conditional expectations coincide with linear projections; the exactness of the recovery operators of Section 5 rests on this fact.

Assumption 2.1 (Stability). *A is Hurwitz: $\max_i \text{Re } \lambda_i(A) < 0$. The state is then mean-square stable with stationary variance $\Sigma_\infty = \int_0^\infty e^{As} \Omega e^{A's} ds$.*

Assumption 2.1 is convenient rather than necessary; the filter-existence results of Section 4 require only detectability and stabilizability, stated there.

2.2 Two primitives for a measured flow

A reported flow is the amount accumulated over the window $(t_{k-1}, t_k]$, published as a rate. Two answers exist to the question of what the accumulated amount *is*; the choice between them is a property of the release's statistical construction, made before any filter is written down (Criterion 7.1 states the assignment rule).

Definition 2.2 (Primitive I: integral form). *The flow is the interval integral of the equilibrium intensity,*

$$y_k^{f,I} = \frac{1}{h} \int_{t_{k-1}}^{t_k} C^f x_u du + \sqrt{h} \text{diag}(\eta_f) \xi_k^f, \quad \xi_k^f \sim \mathcal{N}(0, I_{m_f}). \quad (4)$$

The measured functional specifies no diffusion loading of its own; the flow's within-window stochastic content is exhausted by the integrated state innovation. This is the flow treatment

of the exact-discretisation tradition (Bergstrom, 1986; Harvey and Stock, 1993; Zdrozny, 1988; Christensen et al., 2024).

Definition 2.3 (Primitive II: rate form). *The flow is the window increment of a cumulative observable $y_t^f \in \mathbb{R}^{m_f}$ whose instantaneous accrual is the equilibrium intensity. Because the map from state increments to observation increments is linear, the accrual inherits the state’s Brownian increment with loading $C^f B$:*

$$dy_t^f = \underbrace{C^f x_t dt}_{\text{signal drift}} + \underbrace{C^f B dW_t}_{\text{pass-through}}, \quad (5)$$

and the published figure is the accumulated increment, reported as a rate,

$$y_k^f = \frac{1}{h} \int_{t_{k-1}}^{t_k} dy_u^f + \sqrt{h} \text{diag}(\eta_f) \xi_k^f. \quad (6)$$

The two primitives differ in exactly one term, the pass-through $C^f B dW_t$: the structural increments appear contemporaneously inside the rate-form measurement, and do not appear inside the integral-form measurement. Everything downstream traces to this term. The pass-through should be distinguished from the differential of the signal level itself, $d(C^f x_t) = C^f A x_t dt + C^f B dW_t$, which shares the diffusion but whose window increment is a first difference of levels—a change, not a flow.

2.3 Release convention

Measurement error attaches to the *published figure*, not to the accrual: each release—flow rate or stock reading alike—carries one independent draw of standard deviation $\sqrt{h} \eta_j$,

$$y_k^s = C^s x_{t_k} + \sqrt{h} \text{diag}(\eta_s) \xi_k^s, \quad \xi_k^s \sim \mathcal{N}(0, I_{m_s}), \quad (7)$$

with all ξ ’s i.i.d. standard normal, mutually independent, and independent of W . Three features of the convention matter downstream. First, the error is a property of the release (sampling, seasonal adjustment, later revision) rather than of the underlying accrual—the convention native to discrete measurement equations in the exact-discretisation literature (Christensen et al., 2024)—and flow and stock channels are treated symmetrically. Second, the convention covers arbitrary release calendars: when channel j is released at its own interval h_j (mixed sampling), its per-release variance is $h_j \eta_j^2$, so the noise variance rate is frequency-invariant in continuous-time units. Third, for a filter that assimilates the flow rate continuously over a window of length h , the per-release variance $h \eta_j^2$ corresponds to the continuous noise variance rate $(h \eta_j)^2$ (Section 4.2).

2.4 The exact flow decomposition

Proposition 2.4 (Window decomposition of the rate-form flow). *Under (2) and (5)–(6), with $x_{k-1} \equiv x_{t_{k-1}}$,*

$$y_k^f = \underbrace{\frac{1}{h} C^f \Gamma_h x_{k-1}}_{\text{carry-over}} + \underbrace{\frac{1}{h} C^f \eta_k^f}_{\text{path news}} + \underbrace{\frac{1}{h} C^f B \Delta W_k}_{\text{pass-through}} + \underbrace{\sqrt{h} \text{diag}(\eta_f) \xi_k^f}_{\text{release error}}, \quad (8)$$

where $\Delta W_k = \int_{t_{k-1}}^{t_k} dW_u \sim \mathcal{N}(0, hI_q)$ is the window's structural increment and

$$\eta_k^f = \int_0^h \Gamma(h-s) B dW_{t_{k-1}+s} \quad (9)$$

the within-window drift innovation. Under Primitive I the same decomposition holds with the pass-through term deleted. The four terms are mutually uncorrelated except for $\text{Cov}(\eta_k^f, B\Delta W_k) = V_h \Omega$.

Proof. Variation of constants gives $x_{t_{k-1}+s} = e^{As}x_{k-1} + \int_0^s e^{A(s-r)}B dW_{t_{k-1}+r}$. Integrating over $s \in [0, h]$ and applying the stochastic Fubini theorem (Øksendal, 2003) to the double integral, $\int_0^h \int_0^s e^{A(s-r)}B dW_r ds = \int_0^h \left(\int_r^h e^{A(s-r)} ds \right) B dW_r = \int_0^h \Gamma(h-r)B dW_r$, yields $\int_{t_{k-1}}^{t_k} x_u du = \Gamma_h x_{k-1} + \eta_k^f$. Substituting into $\int dy_u^f = C^f \int x_u du + C^f B \Delta W_k$ and dividing by h gives (8). The cross-moment is $\mathbb{E}[\eta_k^f \Delta W_k' B'] = \int_0^h \Gamma(h-s) B B' ds = V_h \Omega$ by the Itô isometry. $\square$

Each term has a distinct downstream role: the carry-over is the part of the window's flow forecastable from conditions at its opening—common to both primitives; the path news is the extra flow produced within the window because shocks moved the state path—the only stochastic flow content under Primitive I, represented exactly by the Van Loan blocks of Section 3; the pass-through is the window's structural increments imprinted on the datum as they arrive—present under Primitive II only, the source of the cross-covariance of Proposition 2.5, and the channel that makes the contemporaneous shock recoverable online (Section 5).

2.5 Collected ABCD form and the cross-covariance

The system (2), (5), (7) is summarised in the ABCD convention of the linear state-space literature (Fernández-Villaverde et al., 2007): state drift A , state diffusion B , observation loading C , and observation shock-loading

$$D = \begin{bmatrix} C^f B \\ 0_{m_s \times q} \end{bmatrix} \in \mathbb{R}^{m \times q}, \quad BD' = \Omega \begin{bmatrix} (C^f)' & 0_{n \times m_s} \end{bmatrix}, \quad (10)$$

where D loads the observation increments on the *same* Brownian motion as B —not on independent noises. Primitive I is exactly the restriction $D^f \equiv C^f B = 0$ in the measurement specification; the stock rows of D are zero because a sampled level has no increment equation.

Proposition 2.5 (Flow–state cross-covariance). *The state innovation and the rate-form flow-observation innovation are cross-correlated at the covariance rate*

$$S \equiv \frac{1}{dt} \text{Cov}(B dW_t, C^f B dW_t) = B(C^f B)' = \Omega (C^f)' \in \mathbb{R}^{n \times m_f}. \quad (11)$$

The release errors, attached to published figures and independent of W , contribute nothing to S . Under Primitive I, $S = 0$ by construction.

The cross-covariance S is the pass-through's fingerprint on the system's second moments, and the organising object of the framework: everything that distinguishes filters in this note—the gain of Section 4.2, the innovation blocks of Sections 4.4 and 4.6, and the

recovery operators of Section 5—traces to how much of S a representation retains, and at what temporal resolution.

For the flow channels, the accumulation is generated by the augmented pair

$$d \begin{bmatrix} x_t \\ y_t^f \end{bmatrix} = \underbrace{\begin{bmatrix} A & 0_{n \times m_f} \\ C^f & 0_{m_f \times m_f} \end{bmatrix}}_{\mathcal{A}} \begin{bmatrix} x_t \\ y_t^f \end{bmatrix} dt + \underbrace{\begin{bmatrix} B \\ C^f B \end{bmatrix}}_{\mathcal{B}} dW_t, \quad (12)$$

whose exact discretisation delivers the discrete representations of Sections 4.4 and 4.6; the integral-form benchmark corresponds to the same $\mathcal{A}$ with the last rows of $\mathcal{B}$ set to zero.

2.6 Stocks versus flows: two shock channels

A stock carries no separate within-window shock term: its measurement (7) specifies no accumulation and no pass-through. It is not, however, silent about the contemporaneous increment. The sampled level contains the state innovation

$$x_{t_k} = e^{Ah} x_{t_{k-1}} + \eta_k^x, \quad \eta_k^x = \int_0^h e^{A(h-s)} B dW_{t_{k-1}+s}, \quad \text{Cov}(\eta_k^x, \Delta W_k) = \Gamma_h B, \quad (13)$$

so the stock conveys the contemporaneous increment *through the level*—conditionally, insofar as the level is precisely observed and the filter’s prior is tight—whereas the flow carries it *unconditionally*, in the datum itself, through the pass-through of (8). For a scalar state, $\text{corr}(\eta_k^x, \Delta W_k) = \sigma \Gamma_h / \sqrt{\Sigma_{\eta^x} h} \rightarrow 1$ as $h \rightarrow 0$. These are the two complementary channels through which the second object of Definition 4.1 is delivered in Section 5: the flow channel dominates when flows are the precise observables, the level channel when the stocks are, and the stacked operator that uses both spans the entire precision range.

3 Exact Discretisation: Window Moments and the Van Loan Computation

The window decomposition of Proposition 2.4 is made operational by its exact second moments; this section computes them. Over an interval of length h the exact transition is $x_k = e^{Ah} x_{k-1} + \eta_k^x$ with the window moments

$$\begin{aligned} \Sigma_{\eta^x} &= \int_0^h e^{As} \Omega e^{A's} ds, & \Sigma_{\eta^f} &= \text{Var}(\eta_k^f) = \int_0^h \Gamma(\tau) \Omega \Gamma(\tau)' d\tau, \\ \Sigma_{\eta^x \eta^f} &= \text{Cov}(\eta_k^x, \eta_k^f) = \int_0^h e^{A\tau} \Omega \Gamma(\tau)' d\tau, \end{aligned} \quad (14)$$

where the second and third integrals follow from (9) and (13) by the Itô isometry (substituting $\tau = h - s$). All three, together with Γ_h , are computed simultaneously and exactly by a single matrix exponential (Van Loan, 1978).

Algorithm 3.1 (Van Loan block computation). *Form the $4n \times 4n$ block matrix and its*

exponential

$$\tilde{P} = \begin{bmatrix} -A & I & 0 & 0 \\ 0 & -A & \Omega & 0 \\ 0 & 0 & A' & I \\ 0 & 0 & 0 & 0 \end{bmatrix}, \quad e^{\tilde{P}h} = \begin{bmatrix} F_1 & G_1 & H_1 & K_1 \\ 0 & F_2 & G_2 & H_2 \\ 0 & 0 & F_3 & G_3 \\ 0 & 0 & 0 & F_4 \end{bmatrix}. \quad (15)$$

Then

$$e^{Ah} = F'_3, \quad \Gamma_h = G'_3, \quad \Sigma_{\eta^x} = F'_3 G_2, \quad \Sigma_{\eta^x \eta^f} = F'_3 H_2, \quad \Sigma_{\eta^f} = F'_3 K_1 + (F'_3 K_1)'. \quad (16)$$

The identities (16) have been verified numerically against direct quadrature of (14) in general (non-normal, multi-shock) systems. In rate units—the units of the published flow—the flow block is divided by h^2 and the cross block by h , consistently with the observation (6). Two expansions used repeatedly downstream: $\Gamma_h = hI + \frac{h^2}{2}A + O(h^3)$ and $\Sigma_{\eta^x \eta^f} = \frac{h^2}{2}\Omega + O(h^3)$, so the cross block in rate units, $\Sigma_{\eta^x \eta^f}/h = \frac{h}{2}\Omega + O(h^2)$, vanishes as $h \rightarrow 0$ —the fraction of the instantaneous cross-covariance retained by the integral-form representation, developed as the temporal-aggregation factor in the companion paper.

4 The Online Criterion and the Four Filters

This section states the standard the framework is built to meet and the filters measured against it: the online tracking criterion (Definition 4.1); the correlated-noise Kalman–Bucy filter with its two-step discrete-sampling algorithm—the operational core of the framework; and the three discrete benchmarks—the exact discrete filter of the integral form (EDT), the naive Euler scheme, and the exact discrete filter of the rate form (EDT⁺)—against which the applications run their comparisons.

4.1 Online real-time tracking: the qualifying criterion

Nowcasting is the estimation of the current state of a target variable from information available up to the present instant, updated as each observation arrives (Bańbura et al., 2013). The framework of this note is built to a stronger standard, stated once, here, as the qualifying criterion that every filter and operator of the note is measured against, and that the companion applications adopt by reference.

Definition 4.1 (Online real-time tracking). *A filter qualifies for online real-time tracking if it delivers, contemporaneously and using only data observed through t_k , estimates of both*

$$\hat{x}_{t_k} \quad (\text{the latent state}) \quad \text{and} \quad \hat{\varepsilon}_{t_k} \quad (\text{the contemporaneous structural innovations}), \quad (17)$$

the second by a fixed linear map, $\hat{\varepsilon}_{t_k} = \mathcal{M} \nu_{t_k}$, applied to the stacked contemporaneous innovation $\nu_{t_k} = [\nu_{t_k}^f; \nu_{t_k}^s]$: a constant (steady-state) operator, no future data, no backward pass.

The criterion strengthens the nowcasting standard in exactly one respect, the second object. A structural innovation postdates the forecaster’s information set and is unpredictable from it, so it is available either contemporaneously—read from the arriving

datum—or retrospectively, from a completed sample; the nowcasting literature, where it offers structural interpretation at all, typically takes the retrospective route (Bańbura et al., 2013). Requiring the increment *online* is what disciplines the representation choices of this note: the measurement primitive of Section 2.2 determines whether the criterion is attainable, and Sections 5 and 6 quantify how well. The operators $\mathcal{M}$ are derived once, in Section 5, and apply identically to any filter’s own innovations.

4.2 The correlated-noise Kalman–Bucy filter

Consider the idealised limit in which the observables are recorded continuously rather than at releases. Write $\tilde{y}_t \in \mathbb{R}^m$ for the continuously observed cumulative record, with increment

$$d\tilde{y}_t = Cx_t dt + D dW_t + dV_t, \quad (18)$$

where D is the observation shock-loading of (10) and V is a release-noise process independent of W ; write $R \succ 0$ for the *total* observation-disturbance variance rate—the pass-through’s own covariance DD' plus the release-noise rate, assembled channel by channel in (21) below—and $\mathcal{Y}_t = \sigma\{\tilde{y}_u : u \leq t\}$ for the record through t . The Kalman–Bucy filter (Kalman and Bucy, 1961), extended to correlated noise (Anderson and Moore, 1979), propagates $\hat{x}_t = \mathbb{E}[x_t | \mathcal{Y}_t]$ through

$$d\hat{x}_t = A\hat{x}_t dt + K_t (d\tilde{y}_t - C\hat{x}_t dt), \quad K_t = (P_t C' + S_\bullet) R^{-1}, \quad (19)$$

where $S_\bullet = [S \ 0_{n \times m_s}] = BD'$ is the cross-covariance block of (10)–(11), and the error covariance obeys the correlated-noise Riccati equation in reduced (output-injection) form,

$$\dot{P}_t = \bar{A}P_t + P_t\bar{A}' + \bar{Q} - P_t C' R^{-1} C P_t, \quad \bar{A} = A - S_\bullet R^{-1} C, \quad \bar{Q} = \Omega - S_\bullet R^{-1} S_\bullet'. \quad (20)$$

When $S_\bullet = 0$ —the textbook independent-noise case—(20) collapses to the standard Riccati equation (Anderson and Moore, 1979). Section 6 establishes that $\bar{Q} \succeq 0$ holds automatically for R so assembled, so the reduction is always a legitimate filtering problem. If (A, C) is detectable and $(\bar{A}, \bar{Q}^{1/2})$ is stabilizable, the Riccati equation converges to the unique stabilizing solution $P_\infty \succeq 0$ of the algebraic equation, with constant gain $K_\infty = (P_\infty C' + S_\bullet) R^{-1}$ (Kalman and Bucy, 1961; Anderson and Moore, 1979).

4.3 Discrete sampling: the two-step algorithm

Observations arrive at the discrete times $\{t_k\}$. How each observation enters must respect its measurement structure: a flow is intrinsically a rate over $(t_{k-1}, t_k]$ and the continuous-time filter is native to it; a stock is a point-in-time reading at t_k , and smearing it as a rate across the interval would misrepresent (and dilute) the observation. The filter therefore processes the two channel types in two steps per interval.

The flow subsystem carries the entire correlated-noise structure with the flow-only matrices

$$\begin{aligned} R^f &= (C^f B)(C^f B)' + \text{diag}((h\eta_f)^2), & S^f &= \Omega (C^f)', \\ \bar{A}^f &= A - S^f (R^f)^{-1} C^f, & \bar{Q}^f &= \Omega - S^f (R^f)^{-1} (S^f)', \end{aligned} \quad (21)$$

where the rank- q outer product in R^f is the pass-through’s own covariance—the flow rows of D in (10)—and the diagonal carries the release noise at its equivalent continuous variance

rate: a per-release draw of variance $h\eta_j^2$, assimilated continuously over a window of length h , corresponds to the variance rate $(h\eta_j)^2$. The steady-state flow gain is $K_\infty^f = (P_\infty^f(C^f)' + S^f)(R^f)^{-1}$ with P_∞^f the stabilizing solution of the flow-only algebraic Riccati equation; for $n = 1$ it is the positive root of the scalar quadratic $((C^f)'(R^f)^{-1}C^f)P^2 - 2\bar{A}^f P - \bar{Q}^f = 0$.

Algorithm 4.2 (Two-step correlated-noise filter at discrete sampling). *Given the posterior $\hat{x}(t_{k-1})$, $P(t_{k-1})$ and the interval's observations (y_k^f, y_k^s) , the published flow rate and stock reading of (6) and (7):*

Step 1. (Continuous flow update.) *Integrate, for $t \in (t_{k-1}, t_k]$,*

$$\frac{d\hat{x}}{dt} = A\hat{x} + K_\infty^f(y_k^f - C^f\hat{x}), \quad \frac{dP}{dt} = \bar{A}^f P + P(\bar{A}^f)' + \bar{Q}^f - P(C^f)'(R^f)^{-1}C^f P, \quad (22)$$

obtaining the pre-stock quantities $\hat{x}(t_k^-)$, $P(t_k^-)$, which incorporate all flow information over the interval including, through K_∞^f , $\bar{A}^f$, $\bar{Q}^f$, the shared-shock cross-covariance S^f .

Step 2. (Discrete stock update at the endpoint.) *With $R^s = h \text{diag}(\eta_s^2)$,*

$$\begin{aligned} S_k &= C^s P(t_k^-)(C^s)' + R^s, & K_k^s &= P(t_k^-)(C^s)'S_k^{-1}, \\ \hat{x}(t_k) &= \hat{x}(t_k^-) + K_k^s(y_k^s - C^s\hat{x}(t_k^-)), & P(t_k) &= (I - K_k^s C^s)P(t_k^-). \end{aligned} \quad (23)$$

Step 1 holds the gain at its steady-state value K_∞^f rather than recomputing $(P(t)(C^f)' + S^f)(R^f)^{-1}$ along the interval; the Riccati propagation supplies the covariance $P(t_k^-)$ that Step 2's stock gain and the recovery moments of Section 5 are formed from. The algorithm requires no state augmentation and no integral accumulators: flows are handled natively by the continuous propagation, the stocks by exact discrete corrections at their observation instants, so no spurious temporal-aggregation penalty is imposed on the point-in-time channels; the correlation structure does the identifying work that augmentation does in the discrete representations below. Because Step 2 is an exact discrete update, the recursion reduces, as the stocks become the dominant precise channels, to the exact discrete Kalman update of a directly observed state—the mechanism of the Anderson–Moore convergence documented in the companion paper. The construction extends to arbitrary release calendars: each channel is assimilated when it arrives, flows over their own windows h_j and stocks at their own instants, with per-release variances $h_j\eta_j^2$ throughout.

4.4 The exact discrete filter of the integral form (EDT)

The exact discrete-time filter of Primitive I augments the state with one accumulator per flow channel, $\mathbf{z}_k = [x_k'; a_k']'$, $a_k \in \mathbb{R}^{m_f}$, built from the lagged state through the within-period operator Γ_h in rate form with the flow loadings folded into the transition:

$$\mathbf{z}_k = \mathbf{A}_z \mathbf{z}_{k-1} + \mathbf{u}_k, \quad \mathbf{A}_z = \begin{bmatrix} e^{Ah} & 0_{n \times m_f} \\ C^f \Gamma_h / h & 0_{m_f \times m_f} \end{bmatrix}, \quad \mathbf{u}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}), \quad (24)$$

$$\mathbf{Q} = \begin{bmatrix} \Sigma_{\eta^x} & (\Sigma_{\eta^x \eta^f} / h)(C^f)' \\ C^f (\Sigma_{\eta^x \eta^f})' / h & C^f (\Sigma_{\eta^f} / h^2)(C^f)' \end{bmatrix}, \quad (25)$$

with the blocks from Algorithm 3.1. Each flow loads the identity on its own accumulator; each stock loads its row of C^s on the contemporaneous state:

$$\mathbf{Y}_k = \mathbf{C}_{\text{obs}} \mathbf{z}_k + \mathbf{v}_k, \quad \mathbf{C}_{\text{obs}} = \begin{bmatrix} 0_{m_f \times n} & I_{m_f} \\ C^s & 0_{m_s \times m_f} \end{bmatrix}, \quad \mathbf{v}_k \sim \mathcal{N}(\mathbf{0}, h \text{diag}(\eta_f^2; \eta_s^2)), \quad (26)$$

with the standard Kalman recursion at the steady-state gain and accumulators reset after each observation. Within its own measurement model (4) this filter is exact and complete: the Van Loan blocks capture the integrated-state content without approximation error at the interval h . Its signature relative to rate-form data is twofold. First, the accumulator is boundary-anchored: generated from x_{k-1} through the deterministic weight Γ_h/h , not dynamically updated within the interval. Second, the blocks (14) integrate the process noise only, so the pass-through's contributions to the flow disturbance are absent from $\mathbf{Q}$: the retained cross block in rate units is $\Sigma_{\eta^x \eta^f}/h = \frac{h}{2}\Omega + O(h^2)$, a vanishing fraction of the instantaneous cross-covariance S . Fine sampling therefore drives this filter to the *uncorrelated*-noise filter, not to the correlated-noise filter of Section 4.2; the exact discrete filter for which the fine-sampling collapse onto the continuous-time filter does hold is EDT⁺ of Section 4.6.

4.5 The naive Euler filter

The Euler filter replaces the exact discretisation with first-order approximations and ignores the stock–flow distinction, treating every observation as a point-in-time reading:

$$\begin{aligned} A_{\text{Euler}} &= I + Ah && (\text{vs. } e^{Ah}), \\ \Sigma_{\eta, \text{Euler}} &= \Omega h && (\text{vs. } \Sigma_{\eta^x}), \\ y_k^f &= C^f x_{t_k} + \text{error} && (\text{vs. (4) or (6)}). \end{aligned} \quad (27)$$

It commits three errors: an $O(h^2)$ transition error, an innovation-covariance error that miscalibrates the gain, and a flow misspecification that omits the temporal-aggregation operator entirely. The first two shrink as the base frequency rises; the third does not, because it is structural misspecification of the observation equation rather than a discretisation error.

4.6 The exact discrete filter of the rate-form process (EDT⁺)

The fourth filter is the exact discrete filter of Primitive II itself, EDT⁺—the instrument against which the two-step filter of Algorithm 4.2 is certified to attain the minimum-MSE ceiling of the release information set, a certification run by simulation in the companion paper. EDT⁺ shares with the EDT of Section 4.4 the augmented state, the measurement equation (26), the accumulator resets, the steady-state gain, and the noise convention; the only change is the object being discretised. Where the EDT discretises the integral-form accumulator $da = C^f x dt$, EDT⁺ discretises the accumulator whose differential is the observed flow increment (5) itself, $da = C^f x dt + C^f B dW$: the augmented pair $(\mathcal{A}, \mathcal{B})$ of (12) rather than its integral-form restriction. The Van Loan computation applied to this pair returns the EDT blocks *plus* the second moments of the pass-through, which by Proposition 5.1 are

$$\Sigma_{\eta^x \eta^f}^+ = \Sigma_{\eta^x \eta^f} + \Gamma_h \Omega = x^+, \quad \Sigma_{\eta^f}^+ = \Sigma_{\eta^f} + h \Omega + V_h \Omega + \Omega V_h' = v^+, \quad (28)$$

Table 1: The four filters compared

Property	CT (§§4.2–4.3)	EDT (§4.4)	Euler (§4.5)	EDT ⁺ (§4.6)
State dimension	n	$n + m_f$	n	$n + m_f$
Transition	Exact	Exact	$I + Ah$	Exact
Flow handling	Continuous ODE	Rate accumulator	Ignored	Rate accumulator
Stock handling	Discrete jump	Augmented vector	Point-in-time	Augmented vector
Within-interval path	Dynamic	Boundary-anchored	None	Boundary-anchored
Pass-through	In gain, instantaneous	Omitted from $\mathbf{Q}$	Omitted	In $\mathbf{Q}^+$, boundary-anchored
Cross-cov. retained	S (full)	$\frac{h}{2}\Omega + O(h^2)$ per loading	Absent	$S(1 + O(h))$ per loading

with $\mathbf{Q}^+$ assembled from these blocks exactly as in (25), in the same rate units. No new machinery is required: the pass-through enters as bookkeeping in the innovation covariance, not as an algorithmic change. By optimality of the conditional expectation, EDT⁺ is the minimum-MSE filter on the release information set of its calendar; its retained cross block in rate units is $\Sigma_{\eta^x \eta^f}^+ / h = \Gamma_h \Omega / h = \Omega(I + O(h))$ per unit loading—the instantaneous cross-covariance at full strength but boundary-anchored, with an $O(h)$ deficit of pure temporal resolution relative to the continuous-time filter. The certification comparisons are made at steady state, where the MMSE property holds; the transient under mismatched initialisations is not ordered by optimality. The certified decompositions—the two-step filter against EDT⁺ isolating the resolution effect, EDT⁺ against EDT isolating the pass-through content—are quantified in the companion paper.

4.7 Summary

Table 1 collects the four filters. Its CT column reports the properties of the continuous-time representation of Sections 4.2–4.3, the discrete-time columns those of the augmented representations of Sections 4.4–4.6.

The filters occupy positions on a single axis: how much of the pass-through each retains, and at what temporal resolution—instantaneously in the gain (CT), as a vanishing fraction through the integrated state (EDT), not at all (Euler), or at full strength but boundary-anchored (EDT⁺). The quantitative development of this axis—the closed-form factors and the recovery-correlation law—is the subject of the companion paper.

5 Online Structural-Shock Recovery and Attainable Ceilings

The filters of Section 4 deliver the first object of Definition 4.1; this section constructs the second. The window’s structural increment is estimated by a fixed projection on the contemporaneous innovation—exact by the Gaussianity of Section 2, with moments assembled from the window blocks of Section 3—and the projection’s attainable correlation has a closed form: the ceiling that measures the information content of each release, not the merit of any filter.

5.1 The projection and its moments

The second object of the criterion (17) is the window increment $\Delta W_k \in \mathbb{R}^q$. Because ΔW_k postdates all data through t_{k-1} , it is independent of $\mathcal{Y}_{k-1}$, and its minimum-variance estimate given data through t_k conditions on the contemporaneous innovation alone:

$$\widehat{\Delta W}_k = \mathbb{E}[\Delta W_k | \mathcal{Y}_k] = c S_\nu^{-1} \nu_k, \quad \hat{\varepsilon}_{t_k} = \widehat{\Delta W}_k / \sqrt{h}, \quad (29)$$

a fixed steady-state operator on the stacked innovation $\nu_k = [\nu_k^f; \nu_k^s]$. At steady state, with P the posterior error covariance of the exact rate-form representation and window-start error $\tilde{e}_{k-1} = x_{k-1} - \hat{x}_{k-1|k-1}$, the innovations are

$$\begin{aligned} \nu_k^f &= \frac{1}{h} C^f (\Gamma_h \tilde{e}_{k-1} + \eta_k^f + B \Delta W_k) + \sqrt{h} \text{diag}(\eta_f) \xi_k^f, \\ \nu_k^s &= C^s (e^{Ah} \tilde{e}_{k-1} + \eta_k^x) + \sqrt{h} \text{diag}(\eta_s) \xi_k^s. \end{aligned} \quad (30)$$

Proposition 5.1 (Recovery moments). *Write $c = [c^f \ c^s] \in \mathbb{R}^{q \times m}$ for the cross-covariances of ΔW_k with (30), and S_ν for the innovation covariance. Then*

$$c^f = \frac{1}{h} B' (V_h + hI)' (C^f)', \quad c^s = B' \Gamma_h' (C^s)', \quad (31)$$

$$\begin{aligned} S_{ff} &= \frac{1}{h^2} C^f (\Gamma_h P \Gamma_h' + v^+) (C^f)' + h \text{diag}(\eta_f^2), \quad v^+ = \Sigma_{\eta^f} + h \Omega + V_h \Omega + \Omega V_h', \\ S_{fs} &= \frac{1}{h} C^f (\Gamma_h P e^{A'h} + (x^+)') (C^s)', \quad x^+ = \Sigma_{\eta^x \eta^f} + \Gamma_h \Omega, \\ S_{ss} &= C^s (e^{Ah} P e^{A'h} + \Sigma_{\eta^x}) (C^s)' + h \text{diag}(\eta_s^2), \end{aligned} \quad (32)$$

where $v^+ = \text{Var}(\eta_k^f + B \Delta W_k)$ and $x^+ = \text{Cov}(\eta_k^x, \eta_k^f + B \Delta W_k)$ are the exact second moments of the rate-form flow disturbance, and the window moments are those of (14). Under Primitive I the same formulas hold with the pass-through contributions deleted: $c_1^f = \frac{1}{h} B' V_h' (C^f)'$, $v_1^+ = \Sigma_{\eta^f}$, $x_1^+ = \Sigma_{\eta^x \eta^f}$.

Proof. All entries are Itô-isometry computations on (9), (13), and $B \Delta W_k$: $\mathbb{E}[\Delta W_k (\eta_k^f)'] = \int_0^h B' \Gamma(h-s)' ds = B' V_h'$; $\mathbb{E}[\Delta W_k (B \Delta W_k)'] = h B'$; $\mathbb{E}[\eta_k^f (B \Delta W_k)'] = V_h \Omega$; $\mathbb{E}[\eta_k^x (B \Delta W_k)'] = \Gamma_h \Omega$; and $\tilde{e}_{k-1}$, the release errors, and the window increments are mutually uncorrelated at steady state. Each identity has additionally been verified by Monte Carlo in a two-state, two-shock system. $\square$

5.2 The operators and their optimality

Every online linear read of the increment is scored by one formula, and one map maximises it.

Lemma 5.2 (Optimal linear read). *Let ν_k be a zero-mean stacked innovation with covariance $S_\nu \succ 0$ and target covariance $c = [c^f \ c^s] \in \mathbb{R}^{q \times m}$ as in Proposition 5.1, and recall $\text{Var}(\Delta W_k) = hI_q$. For any fixed weight vector $g \in \mathbb{R}^m$ and shock component i ,*

$$\text{corr}(g' \nu_k, \Delta W_{k,i}) = \frac{g' c_i}{\sqrt{g' S_\nu g \cdot h}}, \quad c_i \equiv (c_i)' \in \mathbb{R}^m, \quad (33)$$

and the squared correlation is bounded by the generalised Cauchy–Schwarz inequality,

$$(g'c_i)^2 \leq (g'S_\nu g)(c_i'S_\nu^{-1}c_i), \quad (34)$$

with equality if and only if $g \propto S_\nu^{-1}c_i$. The maximal squared correlation is therefore $(c'S_\nu^{-1}c')_{ii}/h$, and by joint Gaussianity the maximiser is not merely the best linear read but the conditional expectation itself, $\mathbb{E}[\Delta W_k \mid \nu_k] = c'S_\nu^{-1}\nu_k$, with conditional covariance $hI_q - c'S_\nu^{-1}c'$.

Proof. The formula (33) is bilinearity of the covariance with $\text{Var}(\Delta W_{k,i}) = h$. For (34), set $a = S_\nu^{1/2}g$ and $b = S_\nu^{-1/2}c_i$; then $g'c_i = a'b$, $g'S_\nu g = \|a\|^2$, $c_i'S_\nu^{-1}c_i = \|b\|^2$, and $(a'b)^2 \leq \|a\|^2\|b\|^2$ with equality iff $a \parallel b$, i.e. $g \propto S_\nu^{-1}c_i$. Under the joint Gaussianity of Section 2, the linear projection coincides with the conditional expectation and the conditional covariance is the Gaussian projection identity. $\square$

Two online estimators follow, one per information set:

$$\mathcal{M}^f \nu_k^f \equiv \frac{c^f S_{ff}^{-1} \nu_k^f}{\sqrt{h}} \quad (\text{flow-only}), \quad \mathcal{M} \nu_k \equiv \frac{c S_\nu^{-1} \nu_k}{\sqrt{h}} \quad (\text{stacked}), \quad (35)$$

each a constant $q \times m_f$ (respectively $q \times m$) operator on the contemporaneous innovation—hence online: no future data, no backward pass—and each the maximiser of Lemma 5.2 on its information set. The flow-only operator generalises the classical correlated-noise recovery of latent Wiener increments from observed signal increments (Nüsken et al., 2019); its instantaneous direction is $(C^f B)'(R^f)^{-1}\nu^f$, the continuous-time limit of (35). The stacked operator adds the stock rows, reading the increments from the levels through (13). Every filter applies the *same* operators to its *own* innovations; filters differ only through the innovations they produce. This contrasts with ex-post recovery from reduced-form residuals via a generalised inverse, as in Christensen et al. (2024), whose Proposition 4 bounds the inversion error at $O_P(1)$; the operators (35) are exact conditional expectations for their information sets, and their residual is the information limit of the window, stated next.

Algorithm 5.3 (Online recovery procedure). *Once, at the steady state of the maintained representation:*

- (i) *assemble the window moments of Proposition 5.1 at the steady posterior P , giving c and S_ν ;*
- (ii) *form the fixed operators (35), $\mathcal{M} = c S_\nu^{-1}/\sqrt{h}$ and its flow-only restriction;*
- (iii) *record the ceilings $\text{diag}(c S_\nu^{-1}c')/h$ of Proposition 5.4 as the diagnostic of the information set—computed before any data arrive.*

Then, at each release date t_k :

- (iv) *form the filter’s own stacked innovation ν_k , each channel against that filter’s own prediction of it;*
- (v) *read the increment estimate $\hat{\varepsilon}_{t_k} = \mathcal{M} \nu_k$.*

No future data enter, no backward pass is run, and the operator is never recomputed; filters differ only through the innovations entering (iv).

5.3 Attainable ceilings

Proposition 5.4 (Recovery ceilings). *The conditional covariance of the window increment given any subset $\nu_{(\cdot)}$ of the stacked innovation is $\text{Var}(\Delta W_k \mid \nu_{(\cdot)}) = hI_q - c_{(\cdot)}S_{(\cdot)}^{-1}c'_{(\cdot)}$, and the squared correlation between any online estimate of shock component i and the truth is bounded by the Gaussian projection identities*

$$\begin{aligned}\text{corr}_{\max,i}^2(\text{flow-only}) &= \frac{(c^f S_{ff}^{-1} (c^f)')_{ii}}{h}, \\ \text{corr}_{\max,i}^2(\text{stock-only}) &= \frac{(c^s S_{ss}^{-1} (c^s)')_{ii}}{h}, \\ \text{corr}_{\max,i}^2(\text{stacked}) &= \frac{(c S_{\nu}^{-1} c')_{ii}}{h},\end{aligned}\tag{36}$$

all computable in closed form from (31)–(32) and the steady posterior covariance P of the exact rate-form representation. The operators (35) attain their ceilings by construction: the map is exact for its information set, so the correlation level measures the information content of that set, not method error.

Proof. Each ceiling is Lemma 5.2 restricted to the stated sub-innovation, with S_{ν} and c replaced by the corresponding blocks; the conditional covariance is the Gaussian projection identity of the lemma. Attainment holds because the operators (35) are exactly the maximisers for their information sets. $\square$

Under a periodic release calendar (mixed sampling), the steady state of the exact representation is periodic, and (36) delivers one ceiling per within-period observation position—an identification profile across the calendar; its development, and the documented attainment and complementarity of the two arms across precision configurations, are in the companion paper.

6 Well-Posedness and Proximity to Stochastic Singularity

The ceilings of Section 5 rise as the flows become precise—and precise flows push a many-observable, few-shock system toward stochastic singularity. This section shows the filter is well posed up to that boundary and never beyond it: strictly positive release noise is part of the filter’s definition, and the correlated-noise reduction can never turn indefinite.

A system with more observables than structural shocks, $m > q$, is the canonical stochastically singular configuration: without measurement error the observables would obey exact linear relations and the innovation covariance would lose rank (Komunjer and Ng, 2011; Canova and Sala, 2009). In the present framework well-posedness is intrinsic to the filter rather than assumed away.

Proposition 6.1 (Well-posedness of the correlated-noise reduction). *Let $N = \text{diag}((h\eta_f)^2) \succ 0$ and $R^f = (C^f B)(C^f B)' + N$ as in (21). Then:*

- (i) *$R^f \succ 0$ for any C^f and B whenever $N \succ 0$. In the stochastically singular configuration $m_f > q$ —the case of interest—positive release noise is also necessary: the pass-through outer product has rank at most $q < m_f$, so R^f is singular at $N = 0$.*

(ii) $\bar{Q}^f = \Omega - S^f(R^f)^{-1}(S^f)' \succeq 0$ for any $B, C^f, N \succ 0$ —including rank-deficient Ω and any q : the correlated-noise reduction can never produce an indefinite effective process-noise variance rate when R^f is the correctly assembled total flow-disturbance covariance.

(iii) If $\Omega \succ 0$, then

$$\bar{Q}^f = (\Omega^{-1} + (C^f)' N^{-1} C^f)^{-1} \succ 0, \quad (37)$$

strictly positive definite for all positive release noise, monotonically decreasing as N decreases, and $\bar{Q}^f \rightarrow 0$ as $N \rightarrow 0$ if and only if C^f has full column rank. The singular limit is approached, never crossed: strictly positive release noise is a defining condition of the filter, not a regularisation imposed on it.

Proof. For (i), $R^f \succeq N \succ 0$; conversely, for $m_f > q$ the outer product $(C^f B)(C^f B)'$ has rank at most $q < m_f$, so $N = 0$ leaves R^f singular. For (ii), the pair $(B dW_t, C^f B dW_t + dV_t)$, with V the release-noise process of variance rate N independent of W , has joint covariance rate $\begin{bmatrix} \Omega & S^f \\ (S^f)' & R^f \end{bmatrix} \succeq 0$ as the covariance of a linear map of (dW, dV) ; $\bar{Q}^f$ is its Schur complement with respect to the invertible block R^f , hence positive semidefinite. For (iii), the Sherman–Morrison–Woodbury identity gives $(\Omega^{-1} + (C^f)' N^{-1} C^f)^{-1} = \Omega - \Omega(C^f)'(N + C^f \Omega(C^f)')^{-1} C^f \Omega$, and $N + C^f \Omega(C^f)' = N + (C^f B)(C^f B)' = R^f$, which is (37); positivity, monotonicity in N , and the limit statement follow from the left-hand form. For a scalar state with two flow channels, (37) reduces to $\bar{Q}^f = N_1 N_2 \sigma^2 / (\sigma^2(C_2^2 N_1 + C_1^2 N_2) + N_1 N_2)$, the closed form used in the companion paper. The identity and the semidefiniteness in (ii) have been verified numerically in general multi-shock systems, including rank-deficient Ω . $\square$

Proximity to the singular boundary is malign for likelihood-based parameter estimation—rank-deficient innovation covariances obstruct identification of deep parameters from second moments (Komunjer and Ng, 2011; Canova and Sala, 2009)—but favourable for filtering and recovery *given* the parameters: as the flows become jointly precise they pin the shared increments ever more tightly, the ceilings (36) rise monotonically toward their noise-free limits, and the operators attain them throughout. The evidence for this recovery-versus-estimation asymmetry is in the companion paper. What Proposition 6.1 adds is that no separate positivity check on $\bar{Q}^f$ is ever needed in an application, in any dimension: the genuine multi-shock verification burden lies elsewhere, in Criteria 7.4–7.6 below.

7 Criteria for Assessing a State-Space System

The preceding sections took the system as given; this section states, as verifiable conditions on (A, B, C) and the release structure, when a given system fits the rate-form correlated-noise framework and delivers the online objects of Definition 4.1. The conditions are standard objects of linear filtering and system theory (Kalman and Bucy, 1961; Anderson and Moore, 1979), assembled here in the order an application should check them.

Criterion 7.1 (Primitive assignment). *For each flow release, determine from its statistical construction which primitive it realises: if the published figure is the window increment of a cumulative observable accruing at a state-dependent intensity (income earned, prices paid), the release is rate-form and its row of $D^f = C^f B$ is non-zero; if it is a constructed*

interval integral or average of a level, the release is integral-form and its row of D^f is zero. The assignment is a property of the release, decided before the filter, and each filter should be matched to the primitive of its data. Point-in-time readings are stocks and carry D -rows of zero in either case.

Criterion 7.2 (Observability: the latent state is trackable). *The pair (A, C) is observable: $\mathcal{O} = [C'; (CA)'; \dots; (CA^{n-1})']'$ has full column rank n (detectability suffices for a stable filter). Observability depends on (A, C) alone—not on measurement precision or the cross-covariance, which it is blind to: it would hold identically under $S = 0$. It governs the state, and says nothing about the shocks.*

Criterion 7.3 (Order condition: necessary for shock recovery). *$m \geq q$: at least as many observables as structural shocks. With fewer observables than shocks the structural innovations cannot be recovered from the observable history.*

Criterion 7.4 (Fundamentality: causal, online recovery). *The shock-to-observable map CB has full column rank q , and the implied inverse representation is stable (the continuous-time minimum-phase condition). The map is CB rather than the observation shock-loading D of (10) because the stock rows, whose D -rows vanish, convey the increment through the level channel of Section 2.6 rather than instantaneously; CB is the loading that collects both channels. Non-fundamentality would require future observations, defeating the online property. The condition is the continuous-time counterpart of the invertibility requirement under which a structural model admits a VAR representation in its observables (Fernández-Villaverde et al., 2007); there it is binary—the SVAR route recovers the shocks or it does not—whereas the ceilings (36) remain well-defined without it and grade the attainable recovery of each increment. What the pass-through adds beyond Criterion 7.2 is on the gain side: the non-zero S places the shocks in the flow innovations directly, through the additive gain term of (19), and the level channel (13) places them in the stock innovations conditionally; the ceilings (36) quantify how much of each increment the two channels deliver.*

Criterion 7.5 (Separation: shocks can be told apart). *For $q > 1$, shock-type identification is governed by the conditioning of CB , not its rank alone: a full-rank but ill-conditioned CB pins a combination of the shocks precisely while leaving the individual shocks weakly separated. In a mixed stock-flow system the conditioning may rest on the stock channels rather than on near-collinear flow loadings. Report $\text{cond}(CB)$.*

Criterion 7.6 (Well-posedness of the filter). *Strictly positive release noise on every channel, $N \succ 0$ (Proposition 6.1(i)); and existence of the stabilizing steady state: (A, C) detectable and $(\bar{A}, \bar{Q}^{1/2})$ stabilizable, whereupon the correlated-noise Riccati equation (20) has a unique stabilizing solution $P_\infty \succeq 0$ (Anderson and Moore, 1979). Positivity of $\bar{Q}$ itself never fails (Proposition 6.1(ii)) and need not be checked.*

8 Spectral Degradation under Sampling

The framework closes by delimiting its own claims: the bounds of this section state what *any* filter operating on h -sampled releases can achieve—including the sampled-data continuous-time filter of Algorithm 4.2, which conditions on the same discrete information; they bound the information, not the method. They formalise the sense in which the

update geometry differs between continuous and discrete observation, which the flow arm of the two-step filter exploits. Throughout, $f_x(\omega)$ denotes the spectral density of a stationary signal process and $f_v(\omega)$ that of an additive noise process, with observed spectrum $f_y = f_x + f_v$. The optimal (Wiener) linear filter weights each frequency by its signal-to-noise content, with transfer function and minimum error spectrum

$$H(\omega) = \frac{f_x(\omega)}{f_x(\omega) + f_v(\omega)}, \quad f_e(\omega) = \frac{f_x(\omega) f_v(\omega)}{f_x(\omega) + f_v(\omega)}, \quad (38)$$

the integral $\int f_e(\omega) d\omega$ giving the minimum mean-squared estimation error (Wiener, 1949; Whittle, 1983).

Theorem 8.1 (Spectral degradation under sampling). *Let x_t be sampled at interval h , with $\omega_N = \pi/h$ the Nyquist frequency. The spectrum of the sampled process is the folded sum*

$$f_x^h(\omega) = \sum_{j=-\infty}^{\infty} f_x\left(\omega + \frac{2\pi j}{h}\right), \quad |\omega| \leq \omega_N. \quad (39)$$

Consequently $f_x^h(\omega) \geq f_x(\omega)$ for almost every $|\omega| \leq \omega_N$, with strict inequality wherever the out-of-band power $\sum_{j \neq 0} f_x(\omega + 2\pi j/h)$ is positive. Equality (no aliasing) holds for all ω if and only if x_t is band-limited, $f_x(\omega) = 0$ for $|\omega| > \omega_N$. No process driven by Brownian motion satisfies this condition.

Proof. The folding identity (39) is the Poisson summation / sampling theorem (Shannon, 1949). Non-negativity of each summand gives $f_x^h \geq f_x$, strict wherever any image $f_x(\omega + 2\pi j/h)$, $j \neq 0$, is positive. A Brownian-driven diffusion has rational spectral density decaying as ω^{-2} (or slower), hence $f_x(\omega) > 0$ for all finite ω ; the band-limiting condition fails and the inequality is strict on a set of positive measure. $\square$

Theorem 8.2 (Continuous filtering attains the unfolded bound). *Consider estimating x_t from observations corrupted by additive noise v_t . A filter operating on the continuously observed process conditions on the unfolded spectra (f_x, f_v) and attains the minimum error spectrum $f_e(\omega)$ in (38) for all ω . Any filter operating on the h -sampled observations conditions on the folded spectra (f_x^h, f_v^h) and is bounded below by*

$$f_e^h(\omega) = \frac{f_x^h(\omega) f_v^h(\omega)}{f_x^h(\omega) + f_v^h(\omega)} \geq f_e(\omega), \quad (40)$$

with strict inequality wherever the noise is folded ($f_v^h > f_v$). The integrated gap $\int (f_e^h - f_e) d\omega > 0$ is the irreducible cost of sampling, and is not removed by exact discretisation, which reproduces f_e^h without error but cannot recover f_e .

Proof. The continuous-observation case is the standard Wiener result (Wiener, 1949): with full access to (f_x, f_v) the minimum error spectrum is (38). For sampled observations, the optimal linear filter is the Wiener filter for the sampled spectra, giving f_e^h in (40); this is the best any sampled-data filter can achieve. $f_e = f_x f_v / (f_x + f_v)$ is increasing in each of its arguments, and both spectra fold upward, $f_x^h \geq f_x$ and $f_v^h \geq f_v$ (Theorem 8.1 applied to signal and noise alike), so $f_e^h \geq f_e$, strictly where $f_v^h > f_v$. Exact discretisation yields the correct sampled second moments and hence reproduces f_e^h exactly, but f_e^h is bounded below by the folded spectra and cannot attain f_e . $\square$

9 Concluding Remarks

This note can be concluded by answering what is in a measured flow: carry-over, path news, pass-through, release error—and the contributions of this note follows from taking the third term seriously. The pass-through’s fingerprint $S = \Omega(C^f)'$ turns filtering into a correlated-noise problem (Section 4), makes the contemporaneous increment recoverable online with a computable ceiling (Section 5), remains well posed to the singular boundary (Section 6), is checkable on any given system (Section 7), admits an exact discrete filter (Section 4.6), and respects the information bounds of sampling (Section 8). The machinery is stated here once; Rátfai (2026) prices it in closed form and verifies it by simulation; develops it for mixed-frequency framework, and Parra-Alvarez et al. (2026) deploys it on observed US data.

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